

Low-Cost Initial Test (India, target ≈ USD 10,000)

Does a candidate plasticity-promoting treatment (FGF17 + 7,8-DHF) combined with a structured learning environment raise performance across different cognitive tasks in rats? A staged feasibility study that informs whether a larger experiment is justified

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Document	India initial-test protocol, version 2 (after 3 review rounds of 5 Opus + 2 Sonnet critics: round 1 applied here)
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Status	Not approved by an ethics committee. All reviewer verdicts so far: "approve with changes". Prices are estimates, not quotes.

Executive summary

Question. Does a candidate plasticity-promoting treatment — FGF17 (the factor identified in young cerebrospinal fluid, Iram 2022) plus 7,8-DHF (a putative TrkB/BDNF-pathway agonist whose mechanism is disputed) — combined with a structured enriched learning environment (ELE) raise the performance of healthy young rats across three different cognitive tasks, including one not practised in the ELE (transfer), compared with untreated peers? Assumption to confirm: "crf" in the request was read as CSF.

What this is. A small, staged feasibility, variance and direction study with 64 rats in the main experiment (about 10–11 completers per group) and 102 rats in total. It informs — it does not decide or confirm — whether a larger, powered experiment is justified. It cannot show "intelligence", neuroplasticity or any human benefit.

Staging (new in v2). Phase 0a first tests reagents and engagement in 20 satellite rats (does FGF17 activate ERK signalling; does 7,8-DHF?) before the main cohort is committed. Phase 0b is the behavioural study; Phase 0c is histology. This protects the budget against a non-engaging drug.

Budget (honest). Base estimate ≈ USD 9.9k with no reserve, if prices are as assumed and imports are charged duty and GST; ≈ 8.2k if the institute holds a DSIR customs concession and animal per-diem is low; ≈ 11.3–12.3k in a pessimistic case (section 9). USD 10k is therefore a target, not a guarantee; phases can be stopped to hold the cap.

Long-term goal (defined by the researcher, not tested here): to develop this combined approach toward cognitive gain in people with learning disabilities / altered connectivity and in older adults with Alzheimer's disease. Section 2 states this as a hypothesis with a staged roadmap and the known risks, including that FGF-class mitogenic/angiogenic risk may be a hard barrier for non-disease or paediatric enhancement.

1. Aim, hypotheses and what the data can claim

- **Primary aim (feasibility):** estimate completion rate, variance and direction of an aggregate performance index in a 2x2 of drug package x environment, and establish the assays and reagent activity.
- **Exploratory hypothesis:** D+E (drug package + enrichment) shows a higher index than control peers (C) and than either component alone.
- **Transfer:** performance on Hebb-Williams mazes, a task not directly practised in the ELE (the ELE does contain spatial experience such as tunnels, climbing and wheels, which is declared).
- **Allowed claims:** feasibility, attrition, variance, direction and 95% interval of D+E – C, assay performance and pathway engagement (ERK), descriptive structural markers, for the drug package as a whole.
- **Not allowed:** "smarter", "general intelligence" or a g factor (three tasks at n≈10 give no validated general factor); neuroplasticity (no LTP, spines or causal manipulation); single-agent claims; TrkB-specific claims (pERK shows pathway engagement, not TrkB); human or disease claims; replication of Iram 2022 (aged-mouse rescue; this study asks about enhancement in healthy young adults, which no study found has tested).

2. Long-term goal and translational roadmap (later experiments, not this study)

Recommended wording (adopted from the translational reviewer):

"Our long-term aim is to test whether combining a plasticity-promoting biological treatment (an FGF17-type factor plus a validated TrkB/BDNF-pathway agent) with structured enriched learning can produce measurable cognitive benefit in humans. We pursue two **separate** programmes with distinct biology, models, ages and ethics: (a) neurodevelopmental conditions with altered connectivity (learning disabilities), and (b) age-related cognitive decline and early Alzheimer's disease. We hypothesise that supporting synaptic and myelin plasticity is **one possible contributor** to such benefit; this is a hypothesis to be demonstrated with structural and functional evidence, not an assumption, and it is not addressed by the present study. Prior neurotrophin programmes in Alzheimer's have not shown benefit (including a direct-delivery AAV-NGF trial), brain delivery of protein factors is a major obstacle, and both candidate agents face unresolved questions (human CNS delivery of FGF17; whether 7,8-DHF engages TrkB). Each stage proceeds only on pre-registered target-engagement, structural and safety gates."

Stage	Content	Gate to proceed
0 – this study	0a engagement/reagent validation → 0b behaviour in healthy young rats → 0c histology.	Section 8 gates.
1 – powered rat confirmation	Both sexes; full factorial separating FGF17, 7,8-DHF and enrichment (and exercise); sustained infusion by pump; dose-response; durable vs on-drug effects; necessity (TrkB antagonist; FGFR3 knockdown); synaptic and LTP readouts.	Replicated effect on a pre-registered primary; structural effect.
2 – disease models (separate tracks)	Aged rodents (Iram-type rescue); Alzheimer's models (5xFAD, APP/PS1, TgF344-AD) — amyloid-model rescue has repeatedly failed to translate; neurodevelopmental models at developmentally appropriate ages: Fmr1 knockout (fragile X) and Down-syndrome models (Ts65Dn is being superseded) as separate tracks.	Benefit with engagement and structural evidence.
3 – drug development	Clinically deliverable form (intranasal, intrathecal or engineered/receptor-selective agent — all unproven); a validated TrkB-pathway drug with human-relevant PK; two-species toxicology; FGF-class risks (mitogenesis/tumour promotion, angiogenesis); regulatory plan (CDSCO New Drugs and Clinical Trials Rules 2019; ICMR ethics; US/EU IND).	Delivery solved; clean toxicology.
4 – human studies	Phase 1 safety/PK; biomarkers (MRI myelin water fraction/DTI, EEG, cognitive tests); structured learning input as digital training; Phase 2 in MCI/early AD and, separately, in neurodevelopmental populations (DCGI scrutiny; assent plus proxy consent; independent data-safety monitoring).	Safety and biomarker signal.

Hard barriers that must be stated: no validated way exists to deliver FGF17 to the human brain; the mitogenic and angiogenic risk of an FGF-class CNS agent may never clear a paediatric risk-benefit assessment for a non-fatal indication such as a learning disability, and enhancement in people without disease carries a higher ethical bar than treatment of disease; 7,8-DHF is reported not to activate TrkB reliably in cell assays (Boltaev 2017; Todd 2014; a 2021 review); no human trials of FGF17 or 7,8-DHF were found (verify); healthy-animal enhancement may not transfer to disease or to humans. The FGF23/phosphate effect is not the relevant class risk (that is a different, endocrine FGF).

What this study must deliver to earn Stage 1: reagent activity and pathway engagement (Phase 0a), a direction in the aggregate index with a transfer-task contribution, a structural marker (EdU+/Olig2+ cells) and the separation of drug from enrichment (arms D, E, D+E). Covariation of structure with behaviour cannot be judged at this n and is left to Stage 1.

3. Design and animals

Young adult male Wistar rats (about 8–10 weeks at surgery) from a CPCSEA-registered source (ICAR-IVRI Izatnagar INR 350; NISER INR 300–500 plus transport; NIN's animal unit is being reorganised — confirm supply directly). Males only. Randomised by weight and source litter; **each of the two waves contains 8 rats per arm** (wave is a balanced block); coded syringes/vials; surgeon, injector and behaviour operators blind to drug; the environment arm cannot be fully blinded, so ELE handlers and test-day scorers are different people, all behaviour is video-recorded under coded IDs and scored blind, 20% are re-scored for inter-rater agreement (ICC), and apparatus is cleaned between rats.

Group	Content	Enrolled
Main cohort	4 arms × 16: C (vehicle + standard), E (vehicle + ELE), D (FGF17 + 7,8-DHF + standard), D+E; 2 waves × 8 per arm	64
Engagement satellite – FGF17	Vehicle / 15 min / 30 min after an ICV bolus (4 each); snap-frozen hippocampus and periventricular tissue; pERK/total ERK	12
Engagement satellite – DHF	Vehicle vs 7,8-DHF (4 each), 45 min after the 7th daily dose; pERK/total ERK	8
Partner rats	12 unoperated stooge partners (6 assigned to ELE arenas, 6 to control arenas; never swapped; aggression pre-screened; partner ID recorded)	12
Practice/placement	Cannula and dye practice, injection checks	6
Total		102

Enrolment is 16 per arm because ICV-cannula attrition in rats is commonly 30–40%; with 6.5 weeks of a cannula cap, 51 daily injections, food restriction and swimming, losses could exceed this. The plan assumes about 10–11 completers per arm (40–44), reports dropout by arm and task, and runs a sensitivity analysis on animals that complete all three tasks. Exclusions are pre-specified: dye/histology-confirmed ventricular placement, humane endpoints (cumulative-stress scoring), weight loss >15% below the growth-adjusted target (see section 5), and swim speed <50% of cohort median. Housing: pair-housed with a perforated divider if the IAEC permits, otherwise single-housed, identically in all arms (isolation confound declared).

Power (honest). With N=40–44 completers, an interaction contrast reaches 80% power only for $f \approx 0.44$ – 0.46 , and a pairwise D+E vs C comparison only for $d \approx 1.3$ (unadjusted); power at $f = 0.25$ is 0.34 – 0.37 . A cross-species enrichment meta-analysis reports $g \approx 0.44$ in rodents (an unpublished dissertation; not peer-reviewed), and healthy young rats are near ceiling on simple tasks, so a null result is likely even if a modest true effect exists. The study is analysed by interval estimates; pilot effect sizes must not be used to power the next experiment from the point estimate (Kraemer 2006; Leon 2011).

4. Interventions

Component	Specification
FGF17 (days 0–6)	Recombinant mouse FGF17 (sequence alignment mouse vs rat done before ordering). Order one 25 µg vial first, bioassay it (ERK/proliferation assay), then order the rest so a failed vial costs $\approx$ USD 0.4k, not a full re-order. Chronic guide cannula to the right lateral ventricle (scaled coordinates; dye + histology). Four boluses on days 0, 2, 4, 6: 3 µl of ~ 100 µg/ml (0.3 µg each; 1.2 µg per rat) in aCSF + 0.1% BSA + heparin 10 µg/ml , awake, at 0.5–1 µl/min with the injector left in place 2–3 min; aliquots thawed fresh each session, not left at room temperature. Vehicle boluses in all other rats. <u>Dose is unanchored</u> : Iram 2022 used about a week of continuous infusion and the delivered dose was not retrieved; the Methods (PMC9377328) must be read before IAEC submission, and until then the FGF17 contribution is exploratory. The cannula is capped after day 6 and, if histology timing allows, only the dummy remains.
7,8-DHF	5 mg/kg IP daily, days 0–50 (51 injections; rotated injection sites; a low-DMSO vehicle $\leq 10\%$ with PEG300/Tween or a CMC-Na suspension; solubility checked on the bench first; daily timing fixed, after the day's testing/ELE). Rat precedent exists for 5 mg/kg; cognition data in rats are thin; mechanism disputed.
Enriched learning environment (ELE)	Days 0–50, 1 session/day after testing, in the dark phase under red light, same session duration for all (fixed at 60 min). Standard SOP with photos to be appended before IAEC submission, covering: arena size, object inventory (≥ 3 new objects/day from a fixed list), puzzle devices (latch, string-pull, lever and chew devices on a six-level difficulty ladder; promotion after 3 consecutive successes), running wheels (exercise is part of the package and a declared confound; revolutions logged as covariate), tunnels/climbing, a fixed partner rat. <u>Excluded from the ELE: odour discrimination and digging puzzles</u> (they are the dimensions of the set-shifting task). Rewards are small, fixed in calories and deducted from the daily ration. Engagement minutes are logged. Control arena: same partner, handling and duration, empty.
EdU (structure)	25–50 mg/kg IP on days 2, 4, 6, in the main cohort; EdU click chemistry plus Olig2 on the same sections (no acid denaturation). Dose and detectability are tested in the 6-rat practice group first.

5. Timeline (surgery day –7; day 0 = first bolus; two waves offset by 2 weeks)

Days	Activity
–14 to –7	Arrival, quarantine/acclimatisation (1–2 weeks).
–7 to –1	Cannula surgery with analgesia; recovery; handling and arena habituation.

Days	Activity
0–6	FGF17/vehicle boluses (0, 2, 4, 6); EdU (2, 4, 6); ELE/control sessions and 7,8-DHF/vehicle begin on day 0.
14–21	Gradual food restriction to 85–90% of a growth-curve-adjusted target (not of starting weight). Humane exclusion rule: >15% below that target.
22–35	Task 1: digging-based attentional set-shifting (simple/compound discrimination, intradimensional shift, extradimensional shift). Training to criterion with a 7-day cap; tests staggered ≤ 4 rats/day with two operators (each test day 2–4 h/rat). Score: log trials-to-criterion at the extradimensional shift with the intradimensional shift as covariate.
36–44	Task 2: Hebb-Williams maze (3 days adaptation, then a fixed pre-registered subset of 6 problems, same order for all rats, error zones defined in advance). Score: total errors across the 6 problems.
43–44	Return to ad libitum feeding.
45–50	Task 3 (last; non-appetitive): Morris water maze, 25 ± 1 °C, core temperature before/after, warm rest cage, ≥ 5 min between trials; 4 acquisition days, probe 24 h later (mean proximity to the platform; also quadrant time), visible-platform test after the probe; swim speed and floating logged.
51	Perfusion fixation of the whole main cohort.

A T-maze working-memory task was dropped (three spatial tasks would dominate the index, low precision with ≤ 20 trials, ceiling in young rats). The fixed task order (set-shifting → Hebb-Williams → water maze) is declared; carry-over cannot be separated from task. ELE after testing is a post-training consolidation window, declared. Total in-life ≈ 11 weeks including the wave offset.

6. Endpoints and tissue

Aggregate performance index (primary, descriptive). Per-task z-scores are computed with the grand mean and the **pooled within-arm SD estimated in-sample** (no prior data exist, so the scale is not "fixed a priori"); the whole pipeline (z-scoring → index → contrast) is bootstrapped so that the interval reflects the estimated scale. Each z-score is winsorised at ± 3 , signs are aligned, and the index is the mean of the three. The primary index requires all three tasks; animals missing a task are in a sensitivity analysis (per-task mixed model). The index is also reported without the water maze (spatial overlap) and with wheel revolutions as a covariate; body weight is a covariate. The **inter-task correlation matrix is a primary descriptive output** (does a positive manifold exist at all?). Dropout by arm and task is reported.

Block	Readouts	n
Behaviour	Three tasks above; blinded video scoring.	all completers
Structure (main cohort, perfused)	EdU+/Olig2+ cell density in dentate gyrus/CA1 and corpus callosum (systematic sampling; $n \approx 10$ per arm). MBP immunostaining was dropped (insensitive to small new-myelin fractions).	≈ 10 /arm
Engagement (Phase 0a satellites, snap-frozen)	pERK/total ERK in hippocampus and periventricular tissue — rapid ice-cold dissection (<2 min), snap-freezing, phosphatase inhibitors (NaF, orthovanadate, β -glycerophosphate), fixed post-dose intervals; FGF17 at 15 and 30 min post-bolus; 7,8-DHF at 45 min after the 7th daily dose. Reads pathway engagement, not TrkB specificity (pTrkB dropped: antibody unreliable).	4 per group

7. Statistics and pre-registration

- Register on OSF before surgery; ARRIVE 2.0; flow diagram of enrolled, excluded and completed animals.
- Estimation, not significance testing: linear model of the index with drug, environment and wave (balanced), covariates as above; planned contrasts $D+E - C$, $D+E - E$, $D+E - D$ reported as unadjusted estimates with bootstrap 95% intervals (no Holm adjustment because no significance claims are made; the pairwise detectable effect at $\alpha = .05$ is $d \approx 1.3$). Interaction exploratory.
- Sensitivity analyses: completers of all three tasks; per-task mixed model; excluding the water maze; influence of partner ID (random effect) and wheel running.
- Assay readouts are descriptive.

8. Pre-registered decision rules

Phase	Pass condition
0a – reagents	FGF17 bioassay active on the first vial; mouse-rat alignment documented; 7,8-DHF solution stable at the dosing concentration.
0a – FGF17 engagement	pERK/total ERK in FGF17 satellites ≥ 1.5 -fold vehicle at 15 or 30 min. If not met: do not start the main cohort; test continuous infusion or a higher dose in a small follow-up.
0a – DHF engagement	pERK/total ERK ≥ 1.3 -fold vehicle at 45 min after the 7th dose (pathway engagement only). If not met: the DHF component is reported as "effect of unestablished mechanism"; the main cohort may proceed.

Phase	Pass condition
0b - feasibility	$\geq 60\%$ of the 64 enrolled rats complete all three tasks; cannula placement $\geq 75\%$ correct.
0b - direction	Index difference $D+E - C \geq +0.4$ (per-task-SD units; $\approx +0.55$ index-SD) and $D+E \geq$ both E and D in point estimate. The interval is reported but not gated.
0c - structure and manifold	EdU+/Olig2+ density higher in D+E than C (direction) and the median inter-task correlation > 0 .
Overall "go"	Phase 0a passed AND feasibility AND direction AND at least one of the two 0c conditions.

Operating characteristics of the direction rule (simulation, $n=10$ per arm, three tasks with correlation 0.3, task-SD units): probability of "go" is 8% if there are no effects, 37% for a modest effect (E 0.3, D 0.1, D+E 0.4), 62% for the target effect (E 0.3, D 0.2, D+E 0.6), 90% for a large effect (D+E 1.0), and 39% if enrichment alone is responsible (E 0.5, D 0, D+E 0.5) — so the rule is lenient, not a test of synergy. A "go" justifies the larger study only: full factorial separating FGF17, 7,8-DHF and enrichment, both sexes, pumps, and a sample size set from the pilot variance and a minimally important effect (for example 0.5 SD requires about 64 rats per arm in a two-group comparison). A "no-go" identifies the failure mode (delivery, engagement, ceiling, attrition); it is not evidence that the approach cannot work.

9. India logistics and budget

Item	What the review found
Ethics	Rats are small animals: IAEC approval is sufficient (no central CPCSEA forwarding). IAEC meetings are often 2/year to quarterly; proposals are due ≈ 30 days ahead; ask for an interim review. Neonatal/lifespan work is not part of this protocol.
Imports	FGF17 and antibodies are imported (consolidate into one cold-chain shipment where possible): 6–12 weeks; customs-house-agent fees ($\approx$ INR 5–15k per consignment); dry-ice courier. DSIR concession (Notification 51/96) covers "scientific instruments, apparatus and spare parts" at about 5%; coverage of reagents is unconfirmed and the importing institute itself must hold DSIR registration. Without it, landed cost is about 1.3–1.5 $\times$ list. Duty/GST rates are the reviewers' knowledge, not verified by search.
Equipment time	The budget assumes the lab owns or has free access to: stereotaxic frame, water maze + tracking software, cryostat, fluorescence/confocal microscope and stereology software, and a plate reader/western system. If any is billed hourly this is a budget-breaker; confirm ownership before submission.
Timeline	IAEC 1–6 months; procurement in parallel 2–3 (vial-first strategy adds weeks); Phase 0a 1–2; in-life ≈ 3 (incl. quarantine and wave offset); histology and analysis 2–3. Total about 8–14 months from submission.

Budget (USD; list prices found by search plus reviewer knowledge; to be replaced by quotes):

Line	Base	Notes
Rats: 102 incl. satellites, partners, practice, with transport	0.7k	INR 350–500 each
Housing/care ≈ 70 days (incl. quarantine)	1.4k	per-diem not found; optimistic case 0.7k
FGF17 3 $\times$ 25 μ g (list $\approx$ USD 480/50 μ g), landed	1.0k	R&D; price not found; failed vial costs ≈ 0.4 k
7,8-DHF ≈ 4 g (TCI $\approx$ USD 62/g), landed	0.35k	
Cannulas, dummies, screws, cement	0.5k	
Anaesthesia, analgesia, surgical and perfusion consumables (≈ 70 surgeries/handlings)	0.8k	
ELE arena, wheels, objects, puzzle devices	0.6k	local fabrication carries 18% GST
Behaviour consumables and fabrication (set-shift pots/media/odours, Hebb-Williams maze, pellets)	0.7k	
EdU	0.2k	
Antibodies and kits, landed: Olig2 + secondary, EdU click kit, pERK, total ERK	2.4k	list ≈ 1.8 k; with DSIR 1.8k
IHC/western reagents (membranes, ECL, inhibitors)	0.6k	
Shipping, cold chain, customs-agent fees	0.5k	one consolidated shipment
IAEC fee and miscellaneous	0.1k	
Base total (no reserve)	9.9k	
Scenario	Total	Assumptions
Optimistic	≈ 8.2 k	DSIR concession (antibodies at list 1.8k, one cheap shipment 0.3k, FGF17 0.85k); per-diem low (housing 0.7k)
Base	≈ 9.9 k	as above; no reserve
Pessimistic	≈ 11.3 – 12.3 k	15% overrun plus a second FGF17 order, no DSIR, higher per-diem, hourly equipment charges not included

Phase budgeting (so the cap can be held): Phase 0a engagement + reagents ≈ 1.9 k (satellite rats, FGF17 vial, DHF, ERK antibodies, surgery); Phase 0b behaviour ≈ 5.7 k (main cohort, housing, ELE, behaviour, FGF17 and DHF for 64 rats, cannulas); Phase 0c histology ≈ 2.3 k (EdU, Olig2, kit, reagents). Phase 0a also defines the stop rule: a failed gate ends spending at ≈ 1.9 k.

Cuts if prices run high (in order): reduce enrolment to 14/arm; drop the total-ERK antibody only if phospho/total is replaced by a loading control (weakens the gate); drop Phase 0c (keeps the behavioural question). The western block must not be dropped without losing the engagement gates. **Must not be cut:** dye/histology placement check, FGF17 bioassay and alignment, blinding and randomisation, DHF solubility check, visible-platform control.

10. Validation record for this adaptation

Round	Reviewers	Verdicts	Main issues → fix
Earlier A	3 Opus (design; pharmacology; logistics)	All: approve with changes	Pump ends before testing; pump cost; middle-aged rats; tissue conflict → boluses, young adults, new sourcing.
Earlier C	3 Opus + Haiku	All: approve with changes	Transfer overlap; social confound; composite not a g factor; attrition; timeline; budget.
Translational	1 Opus	Approve with changes	Overclaiming; neurotrophin history; FGF safety; paediatric ethics.
Round 1 of 3	5 Opus (biostatistics; pharmacology; behaviour; plasticity assays; India logistics/budget) + 2 Sonnet (consistency audit; fact check)	All 7: approve with changes	Index scale impossible "a priori" and gate miscalibrated (80% false-stop) → in-sample scale, bootstrapped CI, new gate with simulated operating characteristics; Holm vs intervals; BrdU/Olig2 on adjacent sections impossible → EdU same-section; phospho-western pre-analytics; tissue allocation → whole main cohort perfused, separate satellites; pERK ≠ TrkB; FGF17 engagement timing; set-shift/Hebb-Williams timelines unworkable → 3 tasks, staggered, two operators, subset of 6 problems; food-restriction vs exclusion rule; ELE trained set-shift dimensions → removed; partners shared → assigned; antibody import overhead and equipment charges → budget rebuilt with scenarios and phases; 25 internal inconsistencies fixed; fact corrections (neurotrophin wording, NIN, DSIR text, enrichment effect size source).

11. Items the reviewers still do not like / remaining gaps (after round 1 of 3)

#	Gap	Kind
1	FGF17 delivery: boluses are not the Iram infusion; the dose is unanchored until Iram's Methods are read; four transient boluses may be sub-threshold. A null may reflect delivery.	Cannot fix at this budget; pumps in the larger study
2	Power: only very large effects detectable ($d \approx 1.3$); enrichment effects are small to moderate; healthy rats near ceiling; the direction rule is lenient (8% false-go, 39% if enrichment alone).	Intrinsic to a pilot
3	7,8-DHF may be inert in vivo (TrkB agonism disputed); pERK shows pathway engagement only.	Gate; claims withdrawn if it fails
4	ELE confounds social contact, exercise, puzzles and spatial experience; isolation/divider housing; environment arm cannot be fully blinded.	Declared
5	The index is not a validated general factor; fixed task order; two operators; food restriction interacts with arm; cumulative stress (cannula, 51 injections, restriction, swimming) may exceed assumed attrition.	Declared; sensitivity analyses
6	Prices unverified; equipment hourly charges unknown; DSIR coverage of reagents unconfirmed; budget 8.2–12.3k, base 9.9k with no reserve.	Quotes needed before IAEC submission
7	IAEC calendar and a failed FGF17 vial can add months.	Schedule risk
8	No evidence of neuroplasticity (no LTP/spines/synaptic markers); structure descriptive.	Stage 1
9	Literature checks used search snippets only; several references partial; human-trial and neurotrophin statements to be verified.	Researcher to verify

12. References and verification status

Checks were by search snippets only (PubMed, PMC, Crossref and publisher pages unreachable). "Verified" means consistent across indexed records.

#	Citation	Status	Used for
1	Iram T, Kern F, Kaur A, et al. Young CSF restores oligodendrogenesis and memory in aged mice via Fgf17. <i>Nature</i> 2022;605:509–515. doi:10.1038/s41586-022-04722-0	Verified; methods not accessed	FGF17 rationale
2	Boltaev U, et al. <i>Sci Signal</i> 2017;10(493). doi:10.1126/scisignal.aal1670	Verified	7,8-DHF dispute
3	Todd D, Gowers I, Dowler SJ, et al. <i>PLoS One</i> 2014;9(2):e87923. doi:10.1371/journal.pone.0087923	Citation verified; DHF content via reviews	7,8-DHF dispute
4	<i>Pharmaceuticals</i> 2021;14(8):704 (review: do small molecules activate TrkB like BDNF)	Verified	7,8-DHF dispute
5	Rawlings-Mortimer F, et al. <i>Front Syst Neurosci</i> 2023. doi:10.3389/fnsys.2023.1134594	Partial	7,8-DHF during training in mice
6	van Praag H, Kempermann G, Gage FH. <i>Nat Rev Neurosci</i> 2000;1:191–198. doi:10.1038/35044558	Verified	Enrichment
7	Nithianantharajah J, Hannan AJ. <i>Nat Rev Neurosci</i> 2006;7:697–709. doi:10.1038/nrn1970	Verified	Enrichment
8	Galsworthy MJ, et al. <i>Behav Genet</i> 2005;35:675–692. doi:10.1007/s10519-005-3423-9	Verified	General cognitive ability limits
9	Matzel LD, Han YR, Grossman H, et al. <i>J Neurosci</i> 2003 (general learning ability in mice)	Partial	General cognitive ability limits
10	Birrell JM, Brown VJ. <i>J Neurosci</i> 2000;20(11):4320–4324.	Verified	Set-shifting task
11	Rabinovitch MS, Rosvold HE. <i>Can J Psychol</i> 1951;5(3):122–128.	Partial	Hebb-Williams maze
12	Kraemer HC, et al. <i>Arch Gen Psychiatry</i> 2006;63:484–489. doi:10.1001/archpsyc.63.5.484	Verified	Pilot-study caution
13	Leon AC, Davis LL, Kraemer HC. <i>J Psychiatr Res</i> 2011;45:626–629. doi:10.1016/j.jpsychires.2010.10.008	Verified	Pilot-study caution
14	Percie du Sert N, et al. <i>PLoS Biol</i> 2020;18:e3000410. doi:10.1371/journal.pbio.3000410	Verified	ARRIVE 2.0
15	Enrichment meta-analysis (rodent cognition $g \approx 0.44$, 95% CI 0.25–0.62), Univ. at Albany dissertation (scholarsarchive.library.albany.edu/etd/371); Léger 2014 (duration of enrichment effects in mice)	Partial; dissertation not peer-reviewed	Effect-size expectation

Not found: any study of FGF17/FGF-family treatment improving learning in healthy young rodents; any study combining 7,8-DHF with enrichment; rat ICV FGF17 doses; Iram 2022 delivered dose; Indian prices for Alzet pumps and recombinant FGF17; a modern peer-reviewed validation of the Hebb-Williams maze; per-diem housing cost.